

Research Methodology in CS (CSEG3060_3)

Lecture 2: Report Writing Techniques: types of technical reports, report structure (introduction, methodology, results, discussion, conclusion), referencing styles, citations, and use of visual aids

Unit 4: Technical Writing and Research Proposal Development • Mapped Course Outcome: CO4

Lecture 2: Report Writing Techniques

Course: Research Methodology in Computer Science (CSEG3060_3) **Programme:** B.Tech (Computer Science) **Semester:** V **Unit:** 4 — Technical Writing and Research Proposal Development **Topic:** Report Writing Techniques: Types, Structure, Referencing, Citations, and Visual Aids **Mapped Course Outcome:** CO4 **Estimated Duration:** 60 minutes (lecture plus discussion)

1. Learning Objectives

By the end of this lecture, the student should be able to:

1. Distinguish among the major categories of technical reports and select an appropriate type for a given research or engineering task.
 2. Construct a complete technical report following the canonical Introduction–Methodology–Results–and–Discussion (IMRaD) structure.
 3. Apply at least two academic referencing styles correctly, including in-text citations and bibliographic entries.
 4. Select and design appropriate visual aids (tables, figures, charts, diagrams) to support technical arguments.
 5. Critically review a peer's draft report against objective quality criteria.
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2. Introduction: Why Report Writing Matters in Computer Science

Computer science research is increasingly collaborative, interdisciplinary, and globally distributed. Whether the deliverable is a thesis, conference paper, white paper, internal technical memorandum, or a software release note, the artefact's value depends on how effectively it

communicates its findings. A technically brilliant algorithm with a poorly documented evaluation is, in practical terms, indistinguishable from an undocumented idea. Reporting therefore is not a post-hoc administrative duty; it is an integral part of the research method itself.

Robert Day's *How to Write and Publish a Scientific Paper* frames technical writing as a moral obligation: peer review is impossible without a complete, reproducible account of the method. In computer science, where reproducibility crises around benchmarks, model evaluations, and dataset descriptions are well documented, disciplined reporting is the principal instrument of scientific accountability.

Two foundational assumptions underlie this lecture:

- **The reader is busy.** Most readers will skim before they read. The report's structure must therefore allow rapid extraction of conclusions.
- **The reader is sceptical.** Every claim must be supported, every source traceable, and every figure interpretable without recourse to the text.

These assumptions dictate the structural, stylistic, and citation conventions discussed below.

3. Types of Technical Reports

Technical reports vary according to audience, purpose, formality, and length. A robust typology is essential because each type carries its own conventions regarding tone, citation density, visual aids, and review process.

3.1 Classification by Purpose

Type	Primary Purpose	Typical Audience	Distinguishing Features
Research Report / Technical Report	Document an original investigation	Sponsors, peers, regulators	Full IMRaD; detailed methodology; long reference list
Feasibility Report	Assess viability of an option	Management, clients	Comparison tables; cost-benefit analysis; recommendations
Project Report / Thesis	Document a sustained research effort	Examiners, supervisors	Extended literature review; explicit contributions; oral defence
Progress / Interim Report	Update stakeholders on milestones	Sponsors, supervisors	Period-bounded; Gantt charts; risk logs
Laboratory Report	Record an empirical experiment	Instructors, auditors	Hypothesis-driven; numeric data; strict procedural narration
White Paper	Persuade practitioners to adopt a technology	Industry decision-makers	Marketing tone; business case; architecture diagrams
Technical Memorandum (memo)	Communicate decisions or findings internally	Internal teams	Concise; problem-solution structure; rarely peer reviewed
Compliance / Audit Report	Demonstrate adherence to standards	Regulators, accreditation bodies	Checklist-based; evidence-anchored; declarative prose
Incident / Post-mortem Report	Analyse a system failure or outage	Engineering leadership	Timeline; root cause; corrective action plan

3.2 Classification by Formality

- **Formal reports** — printed, bound, with title page, table of contents, executive summary, and full bibliography. Used for theses, sponsored research, and governmental submissions.
- **Informal reports** — short, often single-section, frequently delivered as memos or short technical notes. Used for day-to-day engineering decisions.

3.3 Classification by Length

- **Short reports** (2–10 pages): trip reports, evaluation summaries, informal memos.
- **Medium reports** (10–30 pages): progress reports, laboratory investigations, internal white papers.
- **Long reports** (30+ pages): final research reports, master's theses, dissertations.

3.4 Selecting the Appropriate Type

A practical decision heuristic:

1. **Who is the audience and what decision must they make?** Use the answer to constrain formality and length.
 2. **Is the primary purpose persuasion, documentation, or evaluation?** Each maps to a different report type.
 3. **Will the artefact be peer reviewed?** If yes, follow the conventions of the venue; if internal, follow the organisation's template.
 4. **What is the archival value?** Long-term artefacts demand higher citation discipline and richer appendices.
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4. Canonical Structure: The IMRaD Model and Beyond

The dominant organisational pattern in modern technical and scientific writing is **IMRaD** — Introduction, Methodology, Results, and Discussion — sometimes extended to **IMRaDC** with a formal Conclusion. In computer science, the ACM and IEEE publication templates follow this convention, as do most theses.

4.1 Front Matter

4.1.1 Title Page

The title page is the report's single most-read page. It should contain:

- A specific, descriptive title (often 8–15 words); avoid abbreviations, colloquialisms, and declarative sentences unless required by convention.
- The author's name(s), institutional affiliation(s), and contact details.
- A report number or version identifier.
- Submission date and any sponsorship acknowledgments.

Poor: "An Interesting Study" **Adequate:** "A Comparative Evaluation of Transformer-Based Models for Code Summarisation on the TLCodeSum Dataset"

4.1.2 Abstract

The abstract is a self-contained micro-representation (150–350 words) of the entire work. It typically follows the **four-element convention**:

- **Background** (1–2 sentences): the domain and gap.
- **Objective** (1 sentence): the specific aim.
- **Methods** (1–2 sentences): how the objective was addressed.
- **Results and conclusion** (2–3 sentences): the principal findings and their significance.

Abstracts are indexed by digital libraries; therefore they must avoid acronyms that are not defined, citations to numbered references, and vague claims.

4.1.3 Table of Contents, List of Figures, List of Tables

Generated automatically in LaTeX or Word. Use dot leaders and right-aligned page numbers. Maintain a depth of three heading levels in the table of contents.

4.2 The IMRaD Body

4.2.1 Introduction

The introduction establishes the **why** of the report. A widely recommended internal structure follows the "funnel" pattern:

1. **Broad context** — situate the problem within a wider field.
2. **Specific problem** — narrow to the precise gap addressed.
3. **Prior work** — summarise the most relevant prior studies and how they fall short.
4. **Research question / hypothesis** — state explicitly.
5. **Contribution** — list the paper's claimed contributions.
6. **Outline** — describe the structure of the remainder of the report.

A useful diagnostic for introductions: if the reader cannot, after reading it, articulate the research question, the introduction has failed.

A typical introduction length in a conference paper is 1–1.5 pages; in a thesis it may extend to 15–25 pages, broken into chapters on motivation, related work, and research objectives.

4.2.2 Methodology (or Methods / Materials and Methods)

The methodology establishes the **how**. In computer science this is sometimes called *Approach*, *Method*, or *Design*. Three principles apply:

1. **Reproducibility.** Another competent researcher should be able to repeat the study from your description. Provide equations, hyperparameters, hardware specifications, software versions, dataset identifiers, and evaluation protocols.
2. **Justification.** Every methodological choice should have a rationale citing prior art or a stated trade-off.
3. **Boundary conditions.** State explicit assumptions (e.g., "all experiments were run on a single NVIDIA A100 GPU with CUDA 12.2").

A common structural sub-template:

- Datasets
- Proposed model / algorithm (with formal definitions and diagrams)
- Experimental setup (baselines, metrics, hyperparameters)
- Evaluation protocol (cross-validation, statistical tests, seeds)
- Implementation details (libraries, compute, runtime)

4.2.3 Results

The results section reports **what was observed** without interpretation. Conjugate every statement of finding with the experiment that produced it.

- Use past tense and avoid evaluative adjectives.
- Present numerical results in tables when precise comparison is needed and in figures when trends or distributions dominate.
- State sample sizes, variances, and significance levels.
- If a result contradicts a hypothesis, report it; do not omit.

A useful pattern: each reported outcome is introduced with a topic sentence, supported with a table/figure, and then quantitatively restated, e.g., "*Table 3 indicates that the proposed model achieves a 7.4 percent improvement in mean F1 across the ten folds ($p < 0.01$, Wilcoxon signed-rank).*"

4.2.4 Discussion

The discussion extracts **meaning** from the results. Recommended components:

1. **Summary of principal findings** — restate the two or three key results.
2. **Interpretation** — explain why each result occurred, anchoring the explanation in theory or prior work.
3. **Comparison with prior work** — explicit alignment or divergence.
4. **Implications** — practical and theoretical consequences.
5. **Limitations** — honest articulation of threats to validity.
6. **Future work** — concrete, testable next steps.

A common error is to repeat the introduction in different words. Avoid this by ensuring each discussion paragraph adds interpretive value not already in the results.

4.2.5 Conclusion

The conclusion is short (often 0.5–1 page) and answers: *what did we learn?* A conventional conclusion is structured as:

- One paragraph recapitulating the motivation.
- One paragraph summarising the principal contribution.
- One paragraph on broader impact or future directions.

Conclusions do not introduce new data or new citations.

4.3 Back Matter

- **References / Bibliography** — see Section 6.
- **Appendices** — proofs, full dataset statistics, code listings, additional tables.
- **Acknowledgements** — funding bodies, advisors, technical contributors.
- **Author contributions and conflict-of-interest disclosures** (mandatory in many venues).

5. Micro-Structure at the Paragraph and Sentence Level

Beyond the macro-structure, technical reports impose sentence-level conventions:

- **Topic sentence first.** Each paragraph opens with a sentence stating its claim; supporting sentences follow.
- **Signposting.** Use phrases such as "*In this section, we...*," "*As a consequence,*" "*By contrast,*" and "*We next evaluate...*" to orient the reader.
- **Concrete nouns, precise verbs.** Avoid weak hedges such as "*a thing,*" "*very,*" "*a lot of.*" Replace with "*the proposed classifier,*" "*increases by approximately 4 percent,*" "*evaluated on*

five datasets."

- **Consistent tense.** Past tense for completed work, present tense for established facts and your own claims ("Table 3 shows...").
 - **Voice.** Computer-science writing traditionally tolerates active and passive; modern style guides (ACM 2017) encourage active voice for clarity, e.g., "*We trained five models*" rather than "*Five models were trained.*"
 - **Definitions on first use.** Every acronym (CNN, LoRA, BLEU) is defined once; every symbol in a displayed equation is defined in the preceding text.
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6. Referencing Styles

Referencing has three components: **citation** (in the text), **bibliography entry** (in the reference list), and **style** (the rules connecting them).

6.1 Why Citation Discipline Matters

- Attributes intellectual credit.
- Strengthens the argument by anchoring claims to prior work.
- Protects against plagiarism accusations.
- Enables readers to verify and build upon the work.

6.2 Major Referencing Styles

6.2.1 APA (American Psychological Association)

APA 7 uses an **author–date** in-text system and a *References* list.

In-text: (*Vaswani et al., 2017*) or (*Vaswani, 2017, p. 12*).

Reference:

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30, 5998–6008.

Domain: psychology, education, the social sciences, and increasingly computer science education.

6.2.2 IEEE (Institute of Electrical and Electronics Engineers)

The IEEE style is a **numbered** system with citations in square brackets and a numbered *References* list, ordered by first citation.

In-text: "... as shown in [5]."

Reference:

[5] A. Vaswani et al., "Attention is all you need," in *Proc. 31st Conf. Neural Inf. Process. Syst. (NIPS)*, Long Beach, CA, USA, 2017, pp. 5998–6008.

Domain: electrical engineering, computer science, IEEE journals and conferences — **the de facto standard in CS.**

6.2.3 ACM (Association for Computing Machinery)

ACM follows a numbered, alphabetic-order reference list. Authors use `\bibliographystyle{acm}` in LaTeX (newer ACM styles use `acmart` class).

Reference (sample ACM):

Shazeer, N. and Parmar, N. and Uszkoreit, J. and Jones, L. and Gomez, A. N. and Kaiser, L. and Polosukhin, I. 2017. Attention is all you need. In *Proceedings of the 31st International Conference on Neural Information Processing Systems (NIPS'17)*. Curran Associates Inc., Red Hook, NY, 6000–6010.

6.2.4 Chicago (Chicago Manual of Style, Turabian)

Two variants:

- **Author–date:** used in the natural and social sciences.
- **Notes and bibliography:** used in the humanities; citations appear as footnotes or endnotes with a full bibliography.

6.2.5 MLA (Modern Language Association)

MLA 9 uses parenthetical *author–page* citations and a *Works Cited* list.

In-text: (Vaswani et al. 12).

Domain: humanities, literature, occasionally linguistics.

6.2.6 Harvard

A generic author–date style with national variants (UK, Australian). Many universities adopt a custom Harvard style.

6.2.7 Vancouver (Numeric, Medical)

Numbered citations with bibliographic entries in a prescribed order, mandatory in biomedical journals.

6.2.8 AMA (American Medical Association) and ACS (American Chemical Society)

AMA is used in medical research; ACS in chemistry, occasionally in computational chemistry.

6.3 Choosing and Staying Consistent

A pragmatic decision rule:

1. **Match the venue.** The IEEE Computer Society publishes a "Manuscript Instructions and Templates" page; the ACM template is similarly specified.
2. **When the venue is unspecified (e.g., a thesis or report),** adopt the style mandated by your institution.
3. **Never mix styles.** Every citation and every bibliography entry must follow one system.

A typology at a glance:

Style	In-text Marker	Reference List	Common in
APA	(Author, year)	References, alphabetical	Psychology, education
IEEE	[n]	References, by first citation	CS, EE
ACM	[n]	References, alphabetical	CS
Chicago (notes)	Footnote	Bibliography	Humanities
MLA	(Author page)	Works Cited	Literature
Vancouver	(n)	References, by first citation	Medicine
Harvard	(Author, year)	Reference List, alphabetical	Multidisciplinary

7. Citations in Detail

7.1 Direct Quotation, Paraphrase, and Summary

- **Direct quotation:** reproduces the original wording, enclosed in quotation marks, with page number. Use sparingly in technical writing.
- **Paraphrase:** restates the idea in your own words, with citation. Preferred over quotation in most CS writing.
- **Summary:** condenses a longer work; cite the source as a whole.

7.2 Narrative vs Parenthetical Citation

- **Narrative:** *Vaswani et al. (2017) demonstrate that...* — emphasises the author.
- **Parenthetical:** *... has been demonstrated (Vaswani et al., 2017).* — emphasises the finding.

Most style guides require 3+ authors to be shortened to the first author plus "et al." in parenthetical usage (APA 7). IEEE and ACM list all authors up to a limit (commonly six) before "et al."

7.3 Components of a Bibliography Entry

A complete bibliographic entry usually contains:

1. **Author(s)** with full first names or initials per the style.
2. **Year of publication.**
3. **Title** of the work (article, chapter, book).
4. **Container** — journal name, conference proceedings, book title.
5. **Volume, issue, page range.**
6. **Publisher, location.**
7. **Identifier** — DOI, arXiv ID, ISBN, URL.

Example with DOI (APA 7):

Devlin, J., Chang, M., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of NAACL-HLT 2019*, 4171–4186.
<https://doi.org/10.18653/v1/N19-1423>

7.4 Common Citation Errors

- **Citation orphans** — entries in the reference list never cited in the text.
- **Citation ghosts** — in-text citations with no corresponding bibliography entry.
- **Inconsistent formatting** — mixed APA and IEEE punctuation within the same list.
- **Citing secondary sources** — citing an author as cited in someone else's work rather than reading the original.
- **Stale URLs** — links to dead pages; use DOIs where available.

7.5 Reference Management Tools

Tool	License	Notable Feature
Zotero	Open source	Browser plugin for one-click capture
Mendeley	Freemium	Integrated PDF reader, institutional groups
EndNote	Commercial	Advanced team collaboration
LaTeX BibTeX / bibtex	Open source	Native to LaTeX; highly portable

A reproducible workflow: maintain a single `.bib` file, generate bibliography at compile time, output a chosen `.bst` style.

8. Visual Aids

Visual aids convert raw data into knowledge and shorten the path between claim and evidence. The technical-report ethos requires that every visual element be **self-sufficient** — interpretable without recourse to the prose.

8.1 When to Use a Table vs a Figure

A useful heuristic:

- **Use a table** when the reader must read precise values, when the data are textually heterogeneous, or when the comparison is between discrete categories.
- **Use a figure** when the reader must apprehend a trend, a distribution, or a spatial relationship.

Edward Tufte's *The Visual Display of Quantitative Information* introduces the **data-ink ratio**: maximise the proportion of ink devoted to actual data and minimise decorative elements — chartjunk — such as heavy gridlines, redundant 3D, and ornate borders.

8.2 Charts and Their Appropriate Use

Chart Type	Best For	Cautions
Line chart	Time series, continuous trends	Avoid with too many series (>5)
Bar chart	Comparing quantities across categories	Sort by value unless time is the axis
Histogram	Distribution of a continuous variable	Specify bin width; do not confuse with bar chart
Box plot	Statistical dispersion across groups	Requires statistical literacy from the audience
Scatter plot	Relation between two continuous variables	Beware overplotting; use alpha or jitter
Heatmap	Matrix-valued data (correlations, confusion matrix)	Choose perceptually uniform colormap (viridis)
Sankey / flow	Resource or information flows	Avoid for non-flow data
Network graph	Relational data	Force-directed layouts can be misleading
Confusion matrix	Classification evaluation	Annotate with raw counts and normalised values

8.3 Tables: Construction Rules

- **Number above the table; title above the number.** *"Table 3: Mean F1 across five folds."*
- **Right-align columns of pure numbers;** left-align text columns.
- **Use aligned decimals.** Columns of 0.1240.124, 0.0780.078, 0.3410.341 should be aligned at the decimal place.
- **Avoid vertical rules.** Use horizontal rules only at the top, beneath header, and at the base — the *booktabs* convention.
- **Units in column headers.** *Accuracy (%)*, *Latency (ms)*.
- **Highlighting best results in bold** is acceptable; red highlighting is discouraged because of colour-vision accessibility.
- **Caption explains:** what is shown, the experimental setting, and any symbol definitions.

8.4 Figures: Construction Rules

- **Numbered sequentially** across the whole document.
- **Caption is descriptive**, not interpretative. *"Figure 4: ROC curves for four models."* not *"Figure 4: Model X is best."*
- **Resolution:** ≥ 300 dpi for print, 150 dpi for screenshots.
- **Vector graphics preferred.** Plots saved as `.pdf` or `.svg` scale cleanly.
- **Colour-blind safe palettes.** Use viridis, cividis, or ColorBrewer, never red-green contrasts alone.
- **Fonts inside figures** should match the body text size (typically 8–10 pt).
- **Avoid** dual y-axes unless the relationship is meaningful and explained.

8.5 Algorithms and Pseudo-code Listings

- Use a structured environment in LaTeX (`algorithm` and `algorithmic`).
- Number every line for cross-reference.
- Include input/output specification and complexity comment.

8.6 Code Listings

- Use a syntax-highlighted monospaced listing (`listings` or `minted` in LaTeX).
- Limit to excerpts that highlight the contribution.
- Specify language and version.

8.7 Diagram Conventions

- **Sequence diagrams** for inter-process communication.
- **UML class diagrams** for object-oriented design.
- **Architecture / block diagrams** for system composition.
- **Entity-relationship diagrams** for databases.
- Stick to standard notation; a non-standard diagram is worse than no diagram.

8.8 Accessibility and Inclusion

- Provide alt text for all figures in digital submissions.
- Do not rely solely on colour to convey meaning.
- Use descriptive figure titles that read meaningfully even out of context.

9. Worked Example: A Mini Report Outline for a CS Project

Suppose a student is preparing a final-year report on "*Energy-Efficient Inference Scheduling on Edge Devices*." The structure would be:

1. **Title Page:** Project title, student, supervisor, programme, date.
2. **Abstract:** Background, objective, methods (Raspberry Pi benchmarks), result (15% energy reduction), significance.
3. **Table of Contents.**
4. **Introduction:** growth of edge computing, energy challenge, prior scheduling literature, stated gap.
5. **Methodology:** hardware specs, baseline and proposed scheduler, energy measurement via INA219 sensor, workload suite.
6. **Results:** tables of energy consumption, latency, throughput; figures of latency-vs-energy Pareto fronts.
7. **Discussion:** trade-off interpretation, comparison to Round-Robin baseline, limitations of single-board scope.
8. **Conclusion and Future Work.**
9. **References in IEEE style.**
10. **Appendices:** full measurement logs, code, datasheets.

This outline applies the lecture's principles across the entire writing lifecycle.

10. Common Pitfalls in Report Writing

- **Title–content mismatch.** Promises in title and abstract not delivered in body.
 - **Methodology gap.** Insufficient detail to allow replication.
 - **No clear research question.** Even sound methods cannot redeem an aimless study.
 - **Results–discussion confusion.** Commentary buried inside results paragraphs.
 - **Reference list inconsistent** in formatting.
 - **Visual aids with no purpose** — decorative illustrations that do not advance the argument.
 - **Sweeping overgeneralisation** — extrapolating from $n = 10$ experiments.
 - **Plagiarism of structure** — paraphrasing a published paper too closely even with citations.
 - **Spelling and grammar errors** in the abstract, which is the section reviewers read most closely.
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11. Summary

This lecture consolidated the foundational elements of report writing in computer science. The principal takeaways are:

- Technical reports are of several types — research, feasibility, progress, laboratory, white paper, memo, compliance — each with its own conventions.
- The canonical IMRaD structure (Introduction, Methodology, Results, Discussion, Conclusion), combined with rigorous front and back matter, ensures that the report serves both the busy reader and the sceptical one.
- Referencing is the connective tissue of academic discourse; APA, IEEE, and ACM are the most frequent systems in computer science, and the chosen style must be applied consistently throughout.
- Citations must distinguish between direct quotation, paraphrase, and summary, and bibliography entries must contain author, year, title, container, location, and identifier.
- Visual aids — tables and figures — are not decorative; they are evidence. They must be self-sufficient, accessible, and governed by the data-ink principle.

Mastery of these conventions is a prerequisite for the unit's subsequent lectures on research proposals and presentation assessment.

12. Key Vocabulary

Term	Definition
IMRaD	Introduction–Methodology–Results–and–Discussion structure for scientific reports
Bibliographic entry	A complete reference to a published work in a reference list
In-text citation	A short marker within a sentence that points to a bibliographic entry
DOI	Digital Object Identifier — a persistent link to a digital object
Data-ink ratio	Tufte's ratio of data information to total ink used; maximised in clean charts
Front matter	Pages preceding the introduction (title, abstract, contents)
Back matter	Sections following the conclusion (references, appendices)
Chartjunk	Decorative elements in a chart that obscure the data
Booktabs	LaTeX package enforcing three-rule table style

13. Exam Preparation Tips

1. **Memorise the IMRaD sub-structure.** Examiners commonly ask for the role of each section and a sample opening sentence.
2. **Compare and contrast at least two referencing styles** in tabular form (APA, IEEE, ACM). Practise formatting the same source in two styles.
3. **Distinguish citation, paraphrase, and summary**, with examples.
4. **Identify visual-aid violations.** Given a figure or table, list three improvements: caption, units, accessibility, sort order, etc.
5. **Miniature drafting exercise.** Produce a five-sentence abstract for a fictitious CS study in class.

14. Practice Questions

14.1 Short-Answer Questions

1. List any three types of technical reports and one defining feature of each.

2. What is the difference between a *narrative* and a *parenthetical* citation? Provide one example of each in APA style.
3. State the rule for citing a work with three or more authors in APA 7th edition. Does IEEE follow the same rule?
4. Why is a high data-ink ratio important for figures?

14.2 Medium-Answer Questions

1. Construct the IEEE reference for the following: book by Hopcroft, Motwani, and Ullman titled *Introduction to Automata Theory, Languages, and Computation*, third edition, published by Pearson in 2006.
2. The TLCS 2024 conference requires ACM-style citations. Demonstrate the difference between citing a journal article and citing a book in that style with one example each.
3. Redesign the following figure description to comply with chart hygiene principles: "*Pie chart of programming languages used: Python 50 percent, Java 30 percent, C++ 15 percent, other 5 percent.*" Identify at least three concerns and propose remedies.

14.3 Essay-Type Questions

1. You are writing a 25-page progress report on a sponsored machine-learning research project. Produce a section-level outline using the IMRaD convention and justify the order, length, and content of each section.
2. Critically evaluate the choice of IEEE referencing over APA for a computer-science thesis. What are the practical and ethical implications?

14.4 Application Exercises

1. Take a peer-drafted abstract. Identify violations of the background-objective-methods-results-conclusion structure and rewrite in 200 words.
2. Convert the prose description of an experimental setup into a properly formatted table with units, decimal alignment, and booktabs conventions.
3. Given the reference list of a real conference paper, identify all orphan and ghost citations.

15. Suggested Readings and Tools

15.1 Books

- Day, R. A., & Gastel, B. (2012). *How to Write and Publish a Scientific Paper* (7th ed.). Cambridge University Press.

- Strunk, W., & White, E. B. (2007). *The Elements of Style* (4th ed.). Longman.
- Tufte, E. R. (2001). *The Visual Display of Quantitative Information* (2nd ed.). Graphics Press.
- Higham, N. J. (2002). *Accuracy and Stability of Numerical Algorithms* (2nd ed.). SIAM.
- Booth, W. C., Colomb, G. G., Williams, J. M., Bizup, J., & FitzGerald, W. T. (2016). *The Craft of Research* (4th ed.). University of Chicago Press.

15.2 Standards and Style Guides

- *Publication Manual of the American Psychological Association* (7th ed., 2020).
- *The Chicago Manual of Style* (17th ed., 2017).
- IEEE Editorial Style Manual.
- ACM Author Guidelines (acmart class documentation).

15.3 Tools

- **LaTeX** with `biblatex` , `biber` , `booktabs` , `pgfplots` , `algorithm2e` .
- **Reference managers:** Zotero, Mendeley, EndNote.
- **Statistical and plotting:** R + ggplot2, Python + matplotlib/seaborn/plotly.
- **Grammar:** LanguageTool, Grammarly, Writefull.
- **Reproducibility:** Jupyter, RMarkdown, Quarto, Overleaf.

16. Closing Note for the Instructor

Pedagogically, this lecture works best when followed by a workshop where students bring a one-page draft of their own Introduction and Abstract. The workshop should compare drafts against the criteria in Section 10, generating concrete examples that become reference anchors for the remainder of the unit. Subsequent lectures (Proposal Format and Components; Presentation and Review Committee Assessment) build directly on the conventions established here.